

Subcortical volume and language proficiency in bilinguals and monolinguals: A structural MRI study

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ABSTRACT

The current study focused on an understudied but most prominent bilingual population in the U.S. – heritage bilinguals. The current study combined data from eight MRI studies to examine the relationship between language experience and subcortical gray matter volume in 215 heritage Spanish-English bilinguals and 145 English monolinguals, within and between groups. For bilinguals, higher Spanish (L1) proficiency was related to less volume in the bilateral globus pallidus, and higher English (L2) proficiency and earlier English AoA were related to greater volume in the right thalamus, left accumbens, and bilateral globus pallidus. For monolinguals, higher English proficiency was associated with greater volume only in the right pallidum. These results suggest that subcortical gray matter structures are related to the learning of a second language. Future research is encouraged to understand subcortical adaptation in relation to L1 and L2 acquisition from a developmental perspective.

1. Introduction

Research has established that bilingual experience shapes the brain functionally and structurally. While recent studies have increasingly highlighted the effects of bilingualism on cortical structure (Abutalebi, Canini, Della Rosa, Green, & Weekes, 2015; Klein, Mok, Chen, & Watkins, 2014; Olulade et al., 2016), there is still a scarcity of research directly examining the relationship between bilingual language experience and subcortical structures. Of the available studies that reported subcortical findings, comparisons are usually made between bilingual and monolingual participants (Zou, Ding, Abutalebi, Shu, & Peng, 2012; Abutalebi et al., 2013), and few have investigated volumetric changes within bilinguals as language experience increases (Korenar, Treffers-Daller, & Pliatsikas, 2021). Additionally, among those studies exploring subcortical adaptations, the second language (L2) is often perceived as the weaker language. However, bilinguals exhibit significant heterogeneity in terms of first language (L1) and L2 proficiency, as well as language experience, which may contribute to diverse subcortical adaptations among individuals. For instance, extreme cases such as international adoptees—who are relocated from an L1 environment to an L2 environment—sometimes appear to lose the ability to recognize

their L1. Yet, these individuals may remain more sensitive to their native language compared to those with no prior exposure (Choi, Broersma, & Cutler, 2017; Hyltenstam, Bylund, Abrahamsson, & Park, 2009; Norrman & Bylund, 2016; Park, 2015; Rajagopal et al., 2013).

Moreover, few previous studies on subcortical adaptations to bilingual exposure have addressed the bilingual population that is the most prominent in the U.S.: heritage Spanish speakers who also speak the societal language, English. In the United States, 13.2 % of the population five years and over reported speaking Spanish in addition to English (Census, 2020). Individuals who grow up in Spanish-speaking homes in the U.S. and have some degree of proficiency in Spanish are considered *Spanish heritage speakers* (Valdés, 2014). For heritage speakers, the more proficient language is often the L2 (e.g., English) rather than the L1 (e.g., Spanish). This dynamic suggests that the relationship between age of L2 acquisition, proficiency in L1 and L2, and subcortical volume may follow a different pattern than for other populations of bilinguals, whose bilingualism involves an L2 with lower proficiency. To address this gap in the literature, this paper examines subcortical structural differences (1) between monolinguals and bilinguals and (2) within each language group, focusing on language proficiency and age of acquisition, specifically targeting the under-studied yet significant population of

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bilinguals: Spanish heritage speakers in the U.S.

1.1. Subcortical regions of interest

Studies investigating language experience-related subcortical and cerebellar adaptations typically focus on brain regions that functionally support language, such as the caudate, globus pallidus, nucleus accumbens, putamen, and thalamus (Burgaleta et al., 2016). However, the specific regions and patterns of volume changes vary across studies. In general, these structures help integrate and modulate cortical inputs as situational demands change. For bilinguals, these subcortical structures may be critical when language demands change. For example, the Adaptive Control Hypothesis (ACH; Abutalebi and Green, 2007) outlines the different roles of the basal ganglia, cerebellum, and thalamus depending on whether a bilingual is in a single-language, dual-language, or dense code-switching environment. This perspective of the basal ganglia's role in cognitive coordination aligns with the Conditional Routing Model (CRM; Stocco, Lebiere, & Anderson, 2010; Stocco, Yamasaki, Natalenko, & Prat, 2014; Yamasaki, Stocco, Liu, & Prat, 2019), which posits that the basal ganglia nuclei (e.g., caudate, putamen, globus pallidus, nucleus accumbens) modulate cortical inputs to prefrontal regions. The routing operations defined by the basal ganglia facilitate the transfer of signals between cortical regions through basal ganglia functional pathways. In addition, recent functional neuroimaging studies have shown that the cerebellum is engaged in language tasks, including verbal fluency, word stem completion, word or letter generation, and phonological and semantic processing (for a review, see Stoodley, 2012). The Declarative/Procedural Model (DP model) by Ullman (2020) suggests that the cerebellum is involved in error-based learning of rules that underlie the regularities of complex structures (e.g., past tense of regular verbs), while the basal ganglia is involved in grammar learning (Ullman, 2016, 2020). Given their contribution to language processing, the current study will examine bilateral subcortical gray matter volume (GMV) in the caudate, putamen, thalamus, nucleus accumbens, globus pallidus, and cerebellum among heritage bilinguals and monolinguals.

1.2. Subcortical and cerebellar plasticity in language experience

Different models have attempted to describe characteristics of neural changes associated with language learning and the particular features in language experience that contribute to them. The Expansion-Renormalization Model (ERM; Wenger et al., 2017) states that early stages of skill learning are often associated with gray matter volume (GMV) increase (i.e., expansion), after which additional improvements in task performance often are related to GMV decrease (i.e., renormalization). Consistent with the ERM, the Dynamic Restructuring Model (DRM; Pliatsikas, 2020) argues that structural changes vary depending on the amount of second language (L2) practice and exposure. This model proposes a three-stage adaptation highlighting a cortical to subcortical shift as L2 processing becomes more automated. It proposes that the initial exposure to a language primarily causes cortical gray matter changes, followed by an increase in subcortical and cerebellar volume as language processing becomes more experienced (i.e., consolidation of the language until its peak efficiency), such as an increase in GMV of the putamen, globus pallidus, thalamus, and cerebellum. The model does not specify how increased proficiency rather than the immersive experience in a language affects subcortical structures. It is possible, however, that as language skills increase, gray matter undergoes expansion and then plateaus, independent of the L2 age of acquisition (AoA) (Li et al., 2014). Specifically, in the third stage – peak efficiency, the cerebellum may continue to increase in volume while the caudate may renormalize. The putamen, globus pallidus, and thalamus may reach a plateau.

These models are consistent with recent findings regarding dynamic changes in subcortical volume in relation to language experience

(Korenar, Treffers-Daller, & Pliatsikas, 2021). Perhaps heritage bilingual language acquisition follows the three-stage cortical to subcortical adaptation posited by the DRM, such that there will be an increase in subcortical and cerebellar gray matter volume. It is possible that monolinguals also follow this trajectory over development, and by adulthood, greater language proficiency will be associated with an increase and then a plateau or decrease in subcortical GMV, a sign of the more automatic nature of the process in comparison to children first learning a language. In monolinguals, the subcortical and cerebellar areas have been implicated in language processing (Booth et al., 2007; Chan, Ryan, & Bever, 2013; Tettamanti et al., 2005). However, structural changes related to language experience in healthy monolingual adults remain relatively unclear. Since adult monolinguals are shown to have variations in language proficiency in their native language (Pakulak & Neville, 2010), little is known about the manifestations of individual language proficiency and experience of structural changes. Hence, an additional goal of the current study is to explore the relationship between native language proficiency and subcortical structural volume changes within monolinguals. It is possible that by adulthood, monolinguals' brains have reached the consolidation stage based on the ERM (Wenger et al., 2017) and show a plateau pattern in which proficiency is unrelated to subcortical changes.

1.3. Age of acquisition, language proficiency, and subcortical volume in bilinguals

One common way that researchers study changes in brain structure associated with bilingual experience is by examining the relationship between brain structure and L2 AoA and proficiency. In the context of the current paper, AoA represents the starting age for acquiring L2 (Hernandez & Li, 2007), while proficiency refers to language skills in L1 and L2. In many cases, later L2 AoA is associated with lower ultimate L2 proficiency (Birdsong, 1992). In the current study of heritage bilinguals, AoA may be less correlated with L2 proficiency, as heritage bilinguals are immersed in their L2, and often develop quite high L2 proficiency. Conversely, heritage bilinguals may exhibit more variability in their L1 proficiency, which is likely tied to their L2 AoA (e.g., once they became immersed in the L2, their L1 proficiency may have plateaued). This, in turn, reflects the acquisition of L2 as a function of bilingual experience and usage.

Research has shown that bilingual experience is linked to structural changes in specific brain regions, particularly within the basal ganglia (including the caudate, putamen, nucleus accumbens, and globus pallidus), the thalamus, and the cerebellum. For instance, shape analysis revealed significantly expanded subcortical structures for bilinguals compared to monolinguals, localized in bilateral putamen and thalamus, left pallidum, and right caudate (Burgaleta et al., 2016). Moreover, bilateral expansion of the globus pallidus has been observed in immersed bilinguals — those who have active and continuous usage of two languages (Pliatsikas et al., 2017).

Lower L2 proficiency was found to relate to smaller bilateral putamen for bilingual children (Archila-Suerte et al., 2018). Additionally, greater volume in the putamen has been observed in immersed bilinguals compared to monolinguals (Abutalebi et al., 2013), suggesting a relationship between volume increase and L2 proficiency. The volumes of bilateral caudate nuclei and accumbens also exhibit a non-linear, positive correlation with the amount of bilingual experience, with increases followed by decreases, indicating that both experience and proficiency contribute to structural changes (Korenar, Treffers-Daller, & Pliatsikas, 2021). Furthermore, L2 proficiency have also been associated with increased GMV in the cerebellum (Pliatsikas et al., 2014), which relates to the processing of grammatical rules in L2 (Jeong et al., 2007). A meta-analysis of subcortical activation that compared the neural correlates of different linguistic levels for L1 and L2 in bilingual speakers showed highly consistent results in the left anterior lobe of the cerebellum, as well as left thalamus and right caudate head (Sulpizio et al.,

2020). Notably, a higher volume of the right cerebellum has been associated with better interference suppression from the more proficient language, along with functional activation during interference control (Filippi et al., 2011; Filippi et al., 2020). Recent meta-analysis of GMV studies that directly compared bilinguals and monolinguals also showed consistent results in right cerebellum, despite few consistent results across the studies included in their analysis (Danylkiv & Krafnick, 2020).

Based on these previous findings, we expect that heritage bilinguals will demonstrate larger volumes in the basal ganglia, the thalamus, and the cerebellum with a more proficient L2 proficiency, while L2 AoA may also play a role in shaping these brain structures, potentially reflecting the effects of early exposure to the L2.

1.4. Current study

The present study aims to expand our knowledge of language experience-dependent subcortical neuroplasticity among heritage bilinguals and monolinguals by identifying differences in subcortical volumes and evaluating how AoA and L1 and L2 proficiency might explain these neural differences. Our research questions are: (1) Are there monolingual and bilingual differences in GMV in the subcortical regions of interest (ROIs)?; (2) Within monolinguals, how is language (English being L1) proficiency related to GMV?; and (3) Within bilinguals, how are L2 (English) AoA and language proficiencies (Spanish being L1 and English being L2) related to GMV?. We compared 215 heritage bilinguals' and 145 monolinguals' subcortical volume data. This large sample is well-powered to detect potential differences when comparing monolinguals and bilinguals, as well as correlations within each group (Munson & Hernandez, 2019). In addition, our sample of Spanish-English heritage speakers in the U.S. allows us to consider a special pattern of language experience and elucidate how brain structure reflects language skills within bilinguals that are L2 dominant. Finally, language experience is treated as a continuum for considering the heterogeneous nature of bilinguals. Subcortical ROIs include caudate, putamen, thalamus, nucleus accumbens, globus pallidus, and cerebellum. Based on the DRM and previous findings, we hypothesized that (1) bilinguals would have greater subcortical volume than monolinguals, (2) within monolinguals, higher proficiency would be associated with greater subcortical volume, and (3) within heritage bilinguals, higher English proficiency and earlier AoA would be associated with greater subcortical volume for ROIs similar to English for monolinguals, since English is the primarily used language for both groups. Less research has been done on L1 proficiency and GMV in heritage bilinguals. Thus, we do not have specific hypotheses about the relationship between L1 (Spanish) proficiency and subcortical volume, as these are more exploratory in nature.

2. Method

2.1. Participants

The current study analyzed MRI data from eight research projects conducted from 2007 to 2014 in the Laboratory for the Neural Bases of Bilingualism, which included college-aged English monolingual and Spanish-English bilingual participants. All participants reported being right-handed, having normal or corrected-to-normal vision, and having no history of neurological disorders. Participants were excluded if they could not meet the MRI screening safety requirement (e.g., due to metal or electronic medical implant or claustrophobia). Twelve participants were excluded due to issues with brain scan reconstruction ($n = 6$), technical MRI scanning issues ($n = 5$), and brain trauma ($n = 1$). A final group of 360 participants (145 English monolinguals, 215 heritage Spanish-English bilinguals) was included (for more details, please review Clausenius-Kalman, Vaughn, Archila-Suerte, & Hernandez, 2020; Nguyen et al., 2023). Participants were asked to self-rate their English

and Spanish proficiency in four domains (speaking, listening, reading, and writing) on a scale of 1 to 7 (i.e., 1 = no knowledge; 2 = very poor; 3 = fair; 4 = functional; 5 = good; 6 = very good; 7 = like a native speaker). Individuals were considered monolingual if they reported no knowledge of languages other than English and/or had fewer than two years of foreign language classes. Those who rated themselves below or equal to 2 (very poor to no knowledge) on Spanish proficiency were considered monolingual. Such self-assessment suggests that their ability to engage with Spanish is insufficient for meaningful bilingual communication, thereby justifying their exclusion from the bilingual category. Heritage bilingual participants were those who reported Spanish as a first language and reported no more than two years of foreign language (i.e., not English or Spanish) classes. Participants were between 18 and 45 years of age. Bilinguals and monolinguals did not differ in age and gender, and bilingual participants had on average higher language skills in English than in Spanish. Their English language skills were lower than those of monolinguals (see Table 1 for demographic comparisons between bilinguals and monolinguals).

Socioeconomic status (SES) was measured as parental education on a scale of 1–6 (i.e., 1 = Some elementary school or less than elementary school; 2 = Some high school or less than high school; 3 = High school graduate; 4 = Some college; 5 = College graduate; 6 = Advanced degree). Monolinguals, on average, had higher SES than bilinguals (See Table 1 for descriptive statistics). SES was significantly correlated with Spanish proficiency ($r = 0.20$, $p = 0.006$) and English age of acquisition ($r = 0.62$, $p = 0.037$) but was uncorrelated with English proficiency for bilinguals ($r = 0.13$, $p = 0.08$) and monolinguals ($r = 0.08$, $p = 0.43$). It should be noted that only 177 bilinguals and 100 monolinguals reported SES. In addition, SES was uncorrelated to volume at any subcortical region ($p = 0.11 - 0.95$, partial $\eta^2 = 0.0003 - 0.05$) but differed between language groups and thus was included as a covariate only in the model comparing the two language groups.²

2.2. Materials and Procedures

All participants completed informed consent, background information questionnaires, language history questionnaires, language

Table 1
Summary of demographic characteristics.

	Bilinguals ($n = 215$)		Monolinguals ($n = 145$)		Comparison Estimates
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
English Proficiency	73 %	9 %	81 %	7 %	$t(351.88) = -9.47$, $p < 0.001$, $d = 1.01$
Spanish Proficiency	70 %	14 %			^a $t(214) = 3.13$, $p = 0.002$, $d = 0.43$
English AoA	7.44	5.30			
Age	23.17	4.58	23.29	4.92	$t(281.71) = -0.23$, $p = 0.82$, $d = 0.03$
SES	2.65	1.44	4.30	0.98	$t(269.33) = -11.39$, $p < 0.001$, $d = 1.39$
	<i>n</i>		<i>n</i>		
Gender					$\chi^2(2) = 4.99$, $p = 0.08$
Female	138		96		
Male	62		32		
Did not report	15		19		

Note. ^a Comparison between Spanish and English language skills within bilinguals, bilinguals had significantly higher English proficiency than Spanish proficiency. Bold indicates significant comparisons.

² Including SES in the within group analyses greatly reduced the sample size of each group, leading to lower power to detect a possible effect. Thus, analyses were conducted without including SES, as it was unrelated to subcortical volume at any region across the whole sample and within each group.

proficiency assessments, MRI screening, and an MRI session.

2.2.1. Language proficiency measures

Language proficiencies were assessed using several measures, including the Boston Naming Test (BNT; Kaplan, E., Goodglass, H., & Weintraub, S., 1983) and/or the Woodcock-Muñoz Language Survey – Revised (WLS-R) picture vocabulary subtest along with either the passage comprehension subtest or listening comprehension subtest (Woodcock et al., 2010). A total measure of language proficiency was calculated as an average of the proportion correct (total correct/total possible) across different tests (for more details about how the measures are calculated, review (Claussenius-Kalman et al., 2020).

2.2.2. MRI parameters

Anatomical MRI scans were collected at the Center for Advanced Magnetic Resonance Imaging at Baylor College of Medicine in Houston, Texas. All T1-weighted MPAGE scans were acquired using a 3.0 Tesla Magnetom Trio (Siemens, Germany), using the following parameters for all eight studies: repetition time (TR) = 1200 ms; echo time (TE) = 2.66 ms; flip angle (FA) = 12 degrees; voxel size = 0.479 x 0.479 x 1.0 mm; 192 slices. One scan was collected from each participant. Collecting this single scan took 4 min and 30 s.

2.3. Statistics

Preprocessing was completed in Freesurfer image analysis suite (version 5.3.0, <https://surfer.nmr.mgh.harvard.edu/>); for more information, review (Claussenius-Kalman, et al., 2021; Nguyen et al., 2023). Volumetric registration was used for image reconstruction. Then each MR image was quality checked via visual inspection slice-by-slice in sagittal, coronal, and horizontal planes using Freeview. Brain volumes that had errors were manually fixed and run through a reconstruction of white matter and pial segmentation again. Brain volumes deemed accurate by two researchers went through final reconstruction. Subcortical volume was then extracted through FreeSurfer, during which the subcortical segmentation and volumetric measurement of subcortical gray matter structures were performed (Fischl, 2012). Each participant's structural MRI data were registered to an atlas in FreeSurfer, which was automatically used to label the subcortical structures. Gray matter volume was extracted from our key subcortical structures, and all analyses were conducted using R (<https://www.R-project.org/>).

Five two-tailed general linear models were conducted to analyze the data for the current study separately with each subcortical ROI volume in each hemisphere (ROIs: cerebellum, thalamus, caudate, putamen, globus pallidus, and nucleus accumbens) as the outcome variable, and all analyses controlled for intracranial volume (ICV).

(1) Research question 1: Are there monolingual and bilingual differences in GMV in the subcortical ROIs? Model 1 explored monolingual and bilingual differences in subcortical volume, controlling for SES and participants' age.

$$\text{GMV} \sim \text{language group} + \text{SES} + \text{age} + \text{ICV}$$

(2) Research question 2: Within monolinguals, how is language (English being L1) proficiency related to ROIs' GMV? Model 2 examined the relationship between English proficiency and subcortical volume in monolinguals, controlling for age.

$$\text{Monolingual GMV} \sim \text{English proficiency} + \text{age} + \text{ICV}$$

(3) Research question 3: Within bilinguals, how are L2 (English) AoA and language proficiencies (Spanish being L1 and English being L2) related to ROIs' GMV? It must be noted that English proficiency was moderately correlated with English AoA ($r = -0.43$; $p < 0.001$), which can lead to multicollinearity issue. It was challenging to detangle the effect of AoA and proficiency in the bilingual brain with the current sample. Thus, model 3 and 4 explored proficiency and AoA effects separately, while model 5 included all measures in the same model.

- Model 3 examined the relationship between English AoA with subcortical volume in bilinguals without age as a covariate³ (i.e., main effect of AoA).

$$\text{Bilingual GMV} \sim \text{English AoA} + \text{ICV}$$

- Model 4 examined the relationship between English and Spanish proficiency with subcortical volume in bilinguals, controlling for age (i.e., main effects of English proficiency and Spanish proficiency).

$$\text{Bilingual GMV} \sim \text{English proficiency} + \text{Spanish proficiency} + \text{age} + \text{ICV}$$

- Model 5 examined the relationship between English proficiency, Spanish proficiency, and English AoA with subcortical volume in bilinguals, without age as a covariate.

$$\text{Bilingual GMV} \sim \text{English proficiency} + \text{Spanish proficiency} + \text{English AoA} + \text{ICV}$$

We applied FDR corrections to each analysis and reported only significant results at an FDR-corrected alpha of 0.05. All de-identified data, analyses, and full results can be found at <https://osf.io/z63d4/>.

3. Results

3.1. Language group and subcortical volume

We first ran a linear regression examining whether monolinguals and bilinguals differ in subcortical volume, controlling for SES, age, and total ICV. The two groups did not differ in subcortical volume in any region.⁴

3.2. Monolinguals' subcortical volume

Within monolinguals, we ran a linear regression to examine the relationship between English proficiency and subcortical volume, con-

Table 2

Relationships between English proficiency and subcortical volume in monolinguals.

Region	English Proficiency				
	<i>b</i>	<i>SE</i>	β	<i>p</i>	partial η^2
Left Cerebellum	6545.17	6027.49	0.07	0.33	0.09
Right Cerebellum	3146.89	6311.35	0.03	0.62	0.07
Left Thalamus	908.35	601.54	0.08	0.18	0.21
Right Thalamus	1512.86	675.18	0.12	0.09	0.23
Left Caudate	786.92	439.54	0.12	0.13	0.14
Right Caudate	1190.29	446.54	0.17	0.05⁺	0.18
Left Putamen	1448.65	654.76	0.15	0.09	0.14
Right Putamen	1216.31	637.51	0.13	0.12	0.12
Left Accumbens	154.73	88.92	0.13	0.13	0.10
Right Accumbens	37.83	77.00	0.04	0.62	0.05
Left Pallidum	622.61	297.12	0.16	0.10	0.10
Right Pallidum	646.57	209.02	0.23	0.03[*]	0.16

Note. *p* values are FDR corrected. Bold indicate significant findings. * $p < 0.05$. ⁺ $p < 0.10$.

³ Controlling for participant age in this model might imply the effect of duration of language use, which was not a primary question of interest. Thus, models that included AoA did not include participant age as a covariate.

⁴ English proficiency was related to subcortical volume across all participants. We thus ran an exploratory model to explore the interaction between group and English proficiency, controlling for SES, age, and ICV. No main effect or interaction was significant. While English is the dominant language for the bilinguals, it is still an L2, as opposed to English being L1 for monolinguals. This is consistent with our *a priori* analyses plan to explore the effect of English proficiency separately within each language group.

trolling for total ICV and age. There was a significant positive relationship only at the right pallidum (Table 2, Fig. 1), such that those with higher English proficiency had greater volume ($b = 637.81, p = 0.029$, partial $\eta^2 = 0.15$). There was a marginally significant positive relationship at the right caudate ($b = 1190.293, p = 0.052$, partial $\eta^2 = 0.18$).

3.3. Bilinguals' subcortical volume

3.3.1. Age of acquisition

Within bilinguals, we examined the relationship between English AoA and subcortical volume, controlling for total ICV. Those with earlier English AoA had greater volume in the right thalamus, left accumbens, and bilateral pallidus, with a small to medium effect size (partial η^2 ranges from 0.02 to 0.04; Table 3, Fig. 2). There was a marginally significant negative relationship at the right putamen ($p = 0.054$). In the model with participant's age as a covariate (which may reflect L2 length of usage), AoA was uncorrelated with subcortical volume at all ROIs.⁵

3.3.2. Proficiency

Within bilinguals, we examined the unique relationship between English proficiency, Spanish proficiency, and subcortical volume, controlling for age and total ICV. English proficiency positively correlated with volume at several regions, such that those with higher English proficiency had greater volume in the right thalamus, left accumbens, and bilateral pallidus, with a medium to large effect size (partial η^2 ranges from 0.07 to 0.17, effect size guideline: Lipsey & Wilson, 2001). On the other hand, Spanish proficiency negatively correlated with volume at several regions, such that those with higher Spanish proficiency had less volume at bilateral pallidus. Also, Spanish proficiency positively correlated with the right accumbens with a small to medium effect size (partial η^2 ranges from 0.01 to 0.09; Table 4, Fig. 3).

Finally, we explored the unique relationship between English AoA, English proficiency, and Spanish proficiency with subcortical volume, controlling for total ICV. In this model, only the left and right pallidum were negatively correlated with Spanish proficiency with a small effect size (partial η^2 ranges from 0.05 to 0.09; Table 5). There were marginally significant positive relationships at the right thalamus ($p = 0.057$), left nucleus accumbens ($p = 0.057$), and the left pallidum ($p = 0.057$). English AoA was uncorrelated with subcortical volume at all ROIs.

4. Discussion

The current study examined the relationship between language experience and subcortical GMVs across language groups and within

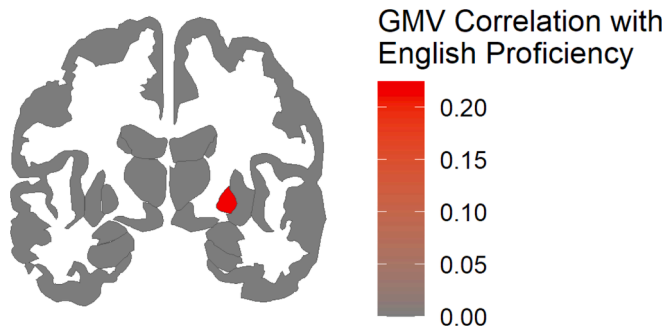


Fig. 1. Subcortical volume correlates with English proficiency in monolinguals. Red indicates positive correlations. The image is a coronal view of the relationships in the brain.

Region	English Age of Acquisition				
	<i>b</i>	<i>SE</i>	β	<i>p</i>	partial η^2
Left Cerebellum	25.36	65.16	0.02	0.84	0.001
Right Cerebellum	-1.70	63.37	-0.001	0.98	0.000
Left Thalamus	-6.34	6.29	-0.05	0.47	0.005
Right Thalamus	-18.87	6.39	-0.14	0.01*	0.040
Left Caudate	0.83	5.20	0.01	0.95	0.000
Right Caudate	-2.37	5.05	-0.03	0.84	0.001
Left Putamen	-9.24	6.34	-0.09	0.29	0.010
Right Putamen	-14.06	6.13	-0.13	0.05⁺	0.024
Left Accumbens	-3.01	0.94	-0.20	0.01*	0.047
Right Accumbens	-0.92	0.81	-0.07	0.45	0.006
Left Pallidum	-9.36	2.93	-0.20	0.01*	0.046
Right Pallidum	-5.26	2.09	-0.15	0.04*	0.029

Note. *p* values are FDR corrected. Bold indicate significant findings. * $p < 0.05$. ⁺ $p < 0.10$.

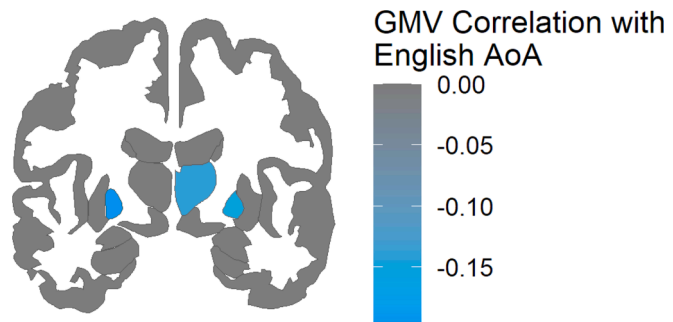


Fig. 2. Subcortical volume correlates with English AoA in heritage bilinguals. Blue indicates negative correlations. The image is a coronal view of the relationships in the brain.

monolinguals and heritage bilinguals, with ROIs including the basal ganglia (caudate, globus pallidus, and nucleus accumbens), thalamus, and cerebellum. While there were no language group differences in subcortical volume, language experience was associated with gray matter differences within groups. We found that (1) regions associated with language expertise differed between heritage bilinguals and monolinguals, (2) within heritage bilinguals, subcortical volumes in language-related regions related to L1 and L2 experience, and (3) the globus pallidus appeared to be related to both monolingual and heritage bilingual language experience across all languages. Specifically, for monolinguals, greater volume in the right pallidum was associated with higher English proficiency, with English being the first and only language. For bilinguals, the opposite pattern of subcortical volume correlation was found between their L1 and L2. Specifically, greater volumes in bilateral pallidus were associated with lower L1 proficiency, with L1 Spanish being their weaker language, and greater volumes in right accumbens were associated with higher L1 proficiency. Additionally, greater volumes in the right thalamus, left accumbens, and bilateral pallidus were associated with higher L2 proficiency, with L2 English being their more proficient language.

4.1. AoA and subcortical volume

For our heritage bilinguals, those with earlier L2 AoA had greater

⁵ Participant's age was significantly correlated with every region.

⁶ We also noticed in our sample that higher Spanish proficiency was associated with higher SES. This could be due to the effects of the quality of inputs. Place and Hoff (2011; 2016) showed a linear relationship between language exposure and vocabulary; higher SES was associated with higher parental education level and, therefore, greater Spanish vocabulary.

Table 4

Relationships between English proficiency and Spanish proficiency with subcortical volume in bilinguals.

Region	English Proficiency					Spanish Proficiency				
	<i>b</i>	<i>SE</i>	β	<i>p</i>	partial η^2	<i>b</i>	<i>SE</i>	β	<i>p</i>	partial η^2
Left Cerebellum	−2172.91	3822.55	−0.03	0.86	0.01	−2898.13	2665.32	−0.06	0.63	0.007
Right Cerebellum	−821.69	3721.01	−0.01	0.90	0.02	−1263.07	2594.52	−0.03	0.68	0.003
Left Thalamus	723.69	363.28	0.10	0.10	0.11	219.24	253.30	0.04	0.63	0.000
Right Thalamus	1255.66	368.32	0.16	0.01*	0.17	−379.33	256.82	−0.07	0.42	0.020
Left Caudate	−110.61	306.01	−0.02	0.86	0.01	145.58	213.37	0.04	0.63	0.002
Right Caudate	176.46	296.72	0.04	0.86	0.03	−143.43	206.89	−0.04	0.63	0.004
Left Putamen	157.85	372.70	0.03	0.86	0.03	18.34	259.87	0.004	0.94	0.001
Right Putamen	698.08	356.22	0.12	0.10	0.07	−186.39	248.38	−0.05	0.63	0.010
Left Accumbens	185.40	53.64	0.21	0.01*	0.11	−23.68	37.40	−0.04	0.63	0.010
Right Accumbens	−3.65	44.72	−0.01	0.94	0.03	96.45	31.18	0.18	0.01*	0.012
Left Pallidum	475.74	164.80	0.18	0.02*	0.07	−472.39	114.91	−0.25	0.001*	0.096
Right Pallidum	298.35	119.78	0.15	0.04*	0.07	−250.05	83.52	−0.18	0.01*	0.052

Note. *p* values are FDR corrected. Bold indicate significant findings. **p* < 0.05.



Fig. 3. Subcortical volume correlates with proficiency in heritage bilinguals. Red indicates positive correlations. Blue indicates negative correlations. The image is a coronal view of the relationships in the brain. The left panel shows English proficiency and the right panel shows Spanish proficiency.

volume in the right thalamus, left accumbens, and bilateral pallidus. L2 AoA was marginally negatively related to the right putamen, such that earlier L2 AoA was associated with a larger volume of the right putamen. Consistent with our finding, the right putamen has been previously shown to be smaller in bilinguals with lower L2 proficiency (Archila-Suerte et al., 2018). Studies have also observed its involvement in language acquisition and processing (Bradley, King, & Hernandez, 2013; Hernandez, 2009; Pliatsikas et al., 2017), which is consistent with our finding of the right putamen's contribution to language acquisition.

4.2. Language proficiency and subcortical volume

We found that greater volumes in the right thalamus, left accumbens, and bilateral pallidus were associated with higher L2 proficiency. Notably, when proficiency and AoA were included in the same model, the relationship between subcortical volume and proficiency persisted, while no relationship was found with AoA. This suggests that the association with proficiency operates independently of AoA. This finding partially aligns with the ERM (Wenger et al., 2017) and contributes to the further development of the DRM (Pliatsikas, 2020). Specifically, as language proficiency increases, the brain regions responsible for language initially expand to incorporate the new skill, subsequently normalizing as this skill becomes more automated—in this case, mastering English as an adult.

Our findings align with existing literature on the dynamic volume adaptation of these brain regions due to bilingual experience. The thalamus, which receives direct projections from the globus pallidus, has previously been implicated in language selection (Abutalebi & Green, 2016), while the nucleus accumbens has been implicated in active language selection and processing (DeLuca et al., 2019). The lateralization of volume change in the left accumbens is also in line with Korenar (2021), which showed that length of language acquisition predicted a reduction in bilateral accumbens, while social language use predicted an

expansion in the left accumbens. The GMV relationships found are consistent with the CRM (Stocco et al., 2010), suggesting the basal ganglia's potential role in actively routing and inhibiting signals during L2 processing.

In our sample of heritage bilinguals, Spanish is the first but weaker language, and the contrasting pattern of GMV correlations with proficiency between L1 and L2 is a novel finding. This pattern mirrors findings from the same sample regarding cortical thickness, which showed a negative correlation with English proficiency in both bilinguals and monolinguals and positively correlated with Spanish proficiency in bilinguals (Nguyen et al., 2023). This aligns with the expansion phase of the ERM (Wenger et al., 2017), where increased proficiency correlates with subcortical GMV expansion. However, current models like the DRM (Pliatsikas, 2020) do not fully account for these results, as they pertain to a first language that is weaker. Our findings may extend the DRM framework (Pliatsikas, 2020), highlighting a cortical-to-subcortical shift in structural adaptation of the more proficient language (L1 for monolinguals, L2 for bilinguals), where the cortex undergoes renormalization while subcortical regions expand.

Notably, the globus pallidus consistently shows volumetric differences associated with language proficiency. We would expect a subcortical to cortical shift of the volume adaptation of the first language, while still observing subcortical volume adaptation for the second language. Prior research has reported increased bilateral pallidus activation in bilinguals during picture naming in L2 and right pallidum activation for L1, suggesting its role in phonological and articulatory properties of language, as well as in a wider network for language control (Liu et al., 2010). However, few studies have examined brain structure or function in relation to L1 proficiency in bilinguals. The globus pallidus internus (GPI) has been theorized to mediate lexical-semantic processes (Crosson, 1985; Whelan et al., 2004). If the globus pallidus plays a role in language immersion and articulation, our results here, alongside existing literature, may indicate its recruitment

Table 5
Relationships between English proficiency, Spanish proficiency, and English AoA with subcortical volume in bilinguals.

Region	English Proficiency					Spanish Proficiency					English AoA				
	b	SE	β	p	partial η^2	b	SE	β	p	partial η^2	b	SE	β	p	partial η^2
Left Cerebellum	-404.67	4401.77	-0.01	1.00	0.01	-4079.60	2883.64	-0.09	0.39	0.01	62.88	81.79	0.05	0.86	0.02
Right Cerebellum	-0.83	4296.92	0.00	1.00	0.02	-2216.53	2814.96	-0.05	0.63	0.00	20.32	79.84	0.02	0.86	0.01
Left Thalamus	726.53	4222.28	0.10	0.21	0.11	117.89	276.64	0.02	0.73	0.00	-2.08	7.85	-0.02	0.86	0.01
Right Thalamus	1130.49	426.64	0.14	0.06 ⁺	0.17	-390.69	279.49	-0.07	0.39	0.02	-6.53	7.93	-0.05	0.86	0.00
Left Caudate	-152.77	352.86	-0.03	0.89	0.01	156.64	231.16	0.04	0.63	0.00	-1.87	6.56	-0.02	0.86	0.00
Right Caudate	221.28	342.29	0.04	0.78	0.03	-191.72	224.24	-0.06	0.63	0.00	1.19	6.36	0.01	0.86	0.01
Left Putamen	-74.19	430.65	-0.01	1.00	0.02	70.74	282.12	0.02	0.80	0.00	-10.50	8.00	-0.10	0.76	0.00
Right Putamen	550.62	414.11	0.09	0.37	0.07	-208.10	271.29	-0.05	0.63	0.01	-7.88	7.69	-0.07	0.86	0.00
Left Accumbens	154.59	62.60	0.18	0.06 ⁺	0.10	-25.95	41.01	-0.04	0.63	0.01	-1.60	1.16	-0.10	0.76	0.00
Right Accumbens	-36.12	54.64	-0.05	0.78	0.02	80.16	35.79	0.15	0.10	0.01	-1.99	1.02	-0.14	0.62	0.01
Left Pallidum	473.62	190.84	0.17	0.06 ⁺	0.07	-504.21	125.02	-0.27	0.001 [*]	0.10	-0.80	3.55	-0.02	0.86	0.00
Right Pallidum	294.26	138.31	0.15	0.10	0.07	-261.49	90.61	-0.19	0.03 [*]	0.05	-0.46	2.57	-0.01	0.86	0.00

Note. p values are FDR corrected. Bold indicate significant findings. * $p < 0.05$. ⁺ $p < 0.10$.

as a result of prolonged language use. We encourage future studies to further explore this topic, focusing on monolingual and bilingual brain structures related to L1 and L2 processing in adulthood. Overall, this current research helps extend the DRM framework (Pliatsikas, 2020) to incorporate language proficiency as a measure of language experience and highlights the role of language dominance in experience-dependent changes within the bilingual brain.

4.3. Limitations and Future Directions

Our current findings revealed a structural correlation with L1 and L2 proficiency and AoA at the nucleus accumbens, globus pallidus, and thalamus, but we found no subcortical differences between monolinguals and bilinguals. Additionally, no significant volume changes were found in the caudate, putamen, or cerebellum for either language group. A possible explanation for these results may be found in the DRM (Pliatsikas, 2020) and ERM (Wenger et al., 2017). These models suggest a trajectory of volume change characterized by an increase followed by a decrease or plateau as language experience changes. Hence, our non-significant findings may reflect a consolidation stage, indicating a plateau in adulthood where proficiency does not correlate with subcortical changes. In this case, perhaps more changes will be observed in early childhood, and future studies are encouraged to adopt a developmental perspective.

Korenar (2021) noted a volumetric trajectory in the bilateral accumbens, suggesting that its volume increases as a function of growing bilingual experiences and plateaus in highly experienced bilinguals. This pattern partially aligns with our findings related to the neighboring caudate nucleus. Given that the role of the nucleus accumbens in bilingualism is less understood, our study helps bridge the gap regarding its potential relationship with bilingual language experience.

Furthermore, specific aspects of different language combination may contribute to the heterogeneity observed in subcortical volume results. For instance, a study that found volume changes in the left putamen (Abutalebi et al., 2013) involved Italian monolinguals and German-Italian-English bilinguals, which presents a different language profile compared to our current sample. A recent meta-analysis of GMV studies comparing bilinguals and monolinguals showed few consistent results across studies, with the exception of findings related to the right cerebellum (Danyliv & Krafnick, 2020). The studies included in the meta-analysis also varied in participants' language profiles, definitions of early versus late bilinguals, thresholds for bilingual proficiency, methodologies, and the MRI processing atlas used, leading to potential heterogeneity in results. Some prior studies also had small sample sizes, which can lead to inflated effect sizes due to inadequate power (Munson & Hernandez, 2019). In contrast, the current study's sample size is a strength, particularly as we focused on heritage speakers, a less studied group. However, the special nature of our sample may limit the generalizability of the findings to other bilingual populations. Future studies should further explore these relationships to better understand the subcortical changes associated with bilingual language experiences. Additionally, while prior research demonstrated the role of subcortical regions in language processing, structural data alone cannot directly explore this. Therefore, future studies should incorporate functional neuroimaging techniques to explore subcortical activity in the current ROIs during language tasks between monolinguals and bilinguals.

Another limitation of this current study is that criteria for grouping participants into monolingual and bilingual categories are somewhat arbitrary, as it relies on self-assessed proficiency, which may not fully capture language capabilities. However, objective language proficiency was used in GMV analysis, which helps ensure that the validity of the results is not compromised. Future studies should consider incorporating additional self-assessment measures to enhance the robustness of language group classification method.

5. Conclusions

The current study examined language background among heritage bilinguals and monolinguals in relation to subcortical volume. It contributes to the existing literature by exploring subcortical volumetric relationships with L1 and L2 proficiency in bilingual heritage speakers compared to monolinguals. The results suggest that relationships between language and GMV changes align within the ERM (Wenger et al., 2017) and extend the DRM (Pliatsikas et al., 2020) by incorporating aspects of proficiency as a proxy for language experience, while also considering the special experience of heritage bilinguals. Specifically, an earlier L2 AoA is associated with greater volume in the right thalamus, left accumbens, and bilateral pallidus among heritage bilinguals. Higher proficiency in L2 (more proficient language) for heritage bilinguals is associated with greater volume in the right thalamus, left accumbens, and bilateral pallidus, while higher proficiency in L1 (less proficient language) is associated with less volume in the bilateral pallidus but more volume in the right accumbens, and these effects are independent of AoA. Additionally, higher proficiency in the only language for monolinguals is associated with greater volume in the right pallidum. These results highlight variability within bilinguals, their difference from monolinguals, and the dynamic nature of language system organization/subcortical representation. Future research is needed to further explore how these relationships evolve across development.

CRediT authorship contribution statement

Yinan Xu: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **My V.H. Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kelly A. Vaughn:** Validation, Supervision, Data curation. **Pilar Archila-Suerte:** Resources, Data curation. **Arturo E. Hernandez:** Validation, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data and code are available at <https://osf.io/z63d4/>

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